MACHINE LEARNING AND DEEP LEARNING

Lab 10: Vision-Language Models (VLMs)

Dr. Giacomo Lucato, a.y. 2025-2026

MODALITIES

- In the context of Deep Learning, data can come to us in different forms, or techinically speaking, **modalities.**
- Examples of modalities are:
  - Vision modalities: images and videos
  - Text: natural language sentences, books, code
  - Audio: speech, music, environmental sounds
  - Structured data: tabular data, spreadsheets
  - 3D/Spatial: point clouds, depth maps
- ML models usually focus on a single modality, but it is possible to construct models able to work across **different modalities.**

VISION-LANGUAGE MODELS (VLM)

- **Vision-Language Models (VLMs)** are a class of multimodal artificial intelligent models designed to simultaneously process, understand, and reason across both **visual data** and **natural language**.
- Most VLMs combine **two powerful neural networks** using a **connector** to blend their outputs:
    - **Vision Encoder:** usually a Vision Transformer (ViT) or a CNN. It extracts features from visual data.
    - **Text Encoder/Decoder:** a Large Language Model (LLM) or a text Transformer. It processes textual prompts or generates natural language responses.
    - **Projection Layer:** a mapping mechanism that translates visual features into visual tokens, making image data looks like text data to the language model.

VISION-LANGUAGE MODELS (VLM)

- VLMs generally fall into two major categories:
  - **Contrastive Models:** they predict which formatting text matches which image out of millions possibilities.
  - **Generative Models:** they take an image and a text prompt, fuse them together, and use an autoregressive decoder to generate brand-new strings of text.

CONTRASTIVE LEARNING IMAGE PRE-TRAINING (CLIP)

- **CLIP** is a VLM falling into the Contrastive Models category.
- It is trained to match images with their corresponding textual captions through a **Contrastive Learning** technique.
- The architecture is based on two cooperating neural networks:
    - **Image Encoder:** takes input images and compresses them into a numerical vector.
    - **Text Encoder:** takes a text caption and compresses it into a vector of the exact same size.

CONTRASTIVE LEARNING IMAGE PRE-TRAINING (CLIP)

- During training, CLIP is fed a batch of N images and N texts simultaneously. This creates a matrix of NxN combinations.
- **Goal:** CLIP tries to maximize **cosine similarity** between the N correct image-text pairs (the diagonal of the matrix).
- **Penalty:** at the same time, it tries to minimize the similarity between all remaining couples.
- By repeating this billions of times, the model pulls the text vector for *"a cute cat"* and the pixel vector of an actual cat into the **exact same neighborhood** in a shared vector space, while pushing the vector for *"a sports car"* far away.

POSSIBLE CLIP APPLICATIONS

- CLIP has several applications, one of these is **image retrieval.**
- Given a large image database, if we want to quickly filter it by some textual prompt, we can use CLIP to retrieve the *top k* images with highest similarity score with the prompt.
- Another possible application is **zero-shot classification:** given a set of images to be classified, and the set of possible (textual) classes, we can use CLIP to compute the similarity score between each possible image-class pair and assign the images the class with highest similarity score.

EXERCISE

- In this exercise you will use a CLIP model to perform some simple tasks, such as Image Retrieval and One-Shot Classification.
- **Task:** download *VLMs.ipynb* and follow the instructions provided within the comments.